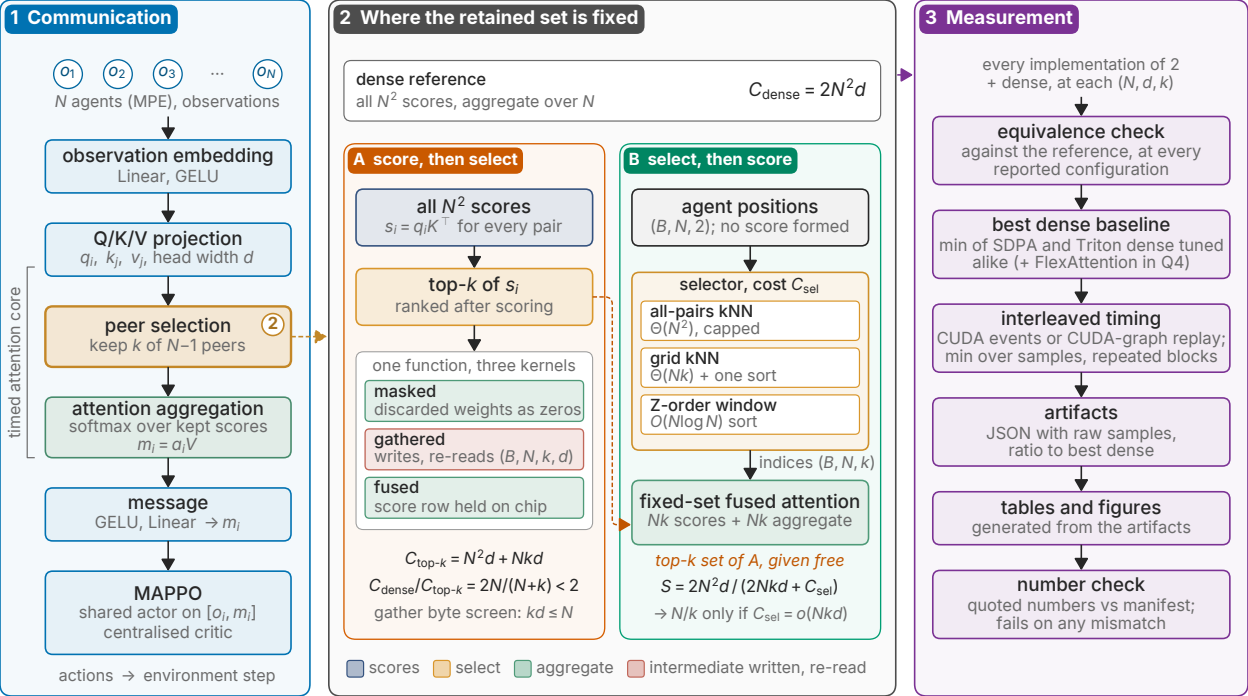

1 Communication
o_1 o_2 o_3 ... o_N
N agents (MPE), observations
observation embedding
Linear, GELU
Q/K/V projection
q_i, k_j, v_j, head width d
timed attention core
peer selection
keep k of $N-1$ peers
2
attention aggregation
softmax over kept scores
$m_i = a_i V$
message
GELU, Linear → m_i
MAPPO
shared actor on $[o_i, m_i]$
centralised critic
actions → environment step
2 Where the retained set is fixed
dense reference
all N^2 scores, aggregate over N
$C_{dense} = 2N^2d$
A score, then select
all N^2 scores
$s_i = q_i K^\top$ for every pair
top-k of s_i
ranked after scoring
one function, three kernels
masked
discarded weights as zeros
gathered
writes, re-reads (B, N, k, d)
fused
score row held on chip
$C_{top\text{-}k} = N^2d + Nkd$
$C_{dense}/C_{top\text{-}k} = 2N/(N+k) < 2$
gather byte screen: $kd \le N$
B select, then score
agent positions
$(B, N, 2)$; no score formed
selector, cost C_{sel}
all-pairs kNN
$\Theta(N^2)$, capped
grid kNN
$\Theta(Nk)$ + one sort
Z-order window
$O(N \log N)$ sort
indices (B, N, k)
fixed-set fused attention
Nk scores + Nk aggregate
top-k set of A, given free
$S = 2N^2d / (2Nkd + C_{sel})$
→ N/k only if $C_{sel} = o(Nkd)$
scores
select
aggregate
intermediate written, re-read
3 Measurement
every implementation of 2 + dense, at each (N, d, k)
equivalence check
against the reference, at every reported configuration
best dense baseline
min of SDPA and Triton dense tuned alike (+ FlexAttention in Q4)
interleaved timing
CUDA events or CUDA-graph replay; min over samples, repeated blocks
artifacts
JSON with raw samples, ratio to best dense
tables and figures
generated from the artifacts
number check
quoted numbers vs manifest; fails on any mismatch